
The Equatorial Ocean

This chapter covers aspects of the equatorial ocean.¹

In midlatitudes there is a fairly clear separation in the time and space scales between balanced and unbalanced motion, and it is useful to recognize this by explicitly filtering out gravity wave motion and considering purely balanced motion, using for example the quasi-geostrophic equations. In equatorial regions, where the Coriolis parameter can become very small and is zero at the equator, the Rossby number may be order unity or larger and such a separation is less useful. However, even as f becomes small, β becomes large and Rossby waves, or their equatorial equivalent, remain as important, or become more even so, than in midlatitudes. The reader may then readily imagine the complications arising even from linear wave problems in equatorial regions: determining the dispersion relation for combined Rossby and gravity waves — Rossby-gravity waves — in a rotating, continuously stratified fluid is an algebraically complex task. The task is greatly simplified by posing the problem in the context of the shallow water equations. The active layer of fluid represents the layer of fluid in and above the main equatorial thermocline, overlying a deep stationary fluid layer of slightly higher density that represents the abyssal ocean. If we accept this physical model we are led to so-called *reduced gravity* equations of motion, as described in chapter 3 of *AOFD*, in which the value of g is replaced by $g' = g\delta\rho/\rho_0$, where $\delta\rho$ is the difference in the value of the density between the two layers of fluid. However, the more accurate equations of motion are the Boussinesq equations. Let us first see how, if the vertical stratification is fixed, the Boussinesq equations may be reduced to the shallow water equations by the device of projecting the equations onto the linear normal modes of the system.

7.1 EQUATIONS OF MOTION

7.1.1 Vertical Normal Modes of the Linear Equations

We begin with the hydrostatic Boussinesq equations, linearized about a state of rest and with fixed stratification, $N(z)$.

$$\frac{\partial u}{\partial t} - fv = -\frac{\partial \phi}{\partial x}, \quad (7.1a)$$

$$\frac{\partial v}{\partial t} + fu = -\frac{\partial \phi}{\partial y}, \quad (7.1b)$$

$$0 = -\frac{\partial \phi}{\partial z} + b, \quad (7.1c)$$

$$\nabla \cdot \mathbf{u} + \frac{\partial w}{\partial z} = 0, \quad (7.1d)$$

$$\frac{\partial b}{\partial t} + wN^2 = 0. \quad (7.1e)$$

These equations are, respectively, the u and v momentum equations, the hydrostatic equation, the mass continuity equation and the buoyancy or thermodynamic equation, with the ∇ operator being purely horizontal. The stratification, $N^2(z)$ is a time unchanging function of z alone. The boundary conditions on these equations are The ‘problem’ with these equations is that there are five independent variables in three spatial coordinates so that even the linear problems are algebraically complex, especially when f is variable. The equations are more general than is needed, because it is often observed that the vertical structure of solutions is relatively simple, especially in linear problems. A solution is to project the vertical structure onto appropriate eigenfunctions, and then to retain a very small number — often only one — of these eigenfunctions.

To determine what those eigenfunctions should be, we combine the hydrostatic and buoyancy equations to give

$$\frac{\partial}{\partial t} \left(\frac{\phi_z}{N^2} \right) + w = 0. \quad (7.2)$$

Differentiating with respect to z and using the mass continuity equation gives

$$\frac{\partial}{\partial t} \left(\frac{\phi_z}{N^2} \right)_z - \nabla \cdot \mathbf{u} = 0. \quad (7.3)$$

It is this equation that motivates our choice of basis functions: we choose to expand the pressure and horizontal components of velocity in terms and an eigenfunction that satisfies the following Sturm–Liouville problem.

$$\frac{d}{dz} \left(\frac{1}{N^2} \frac{dC_m}{dz} \right) + \frac{1}{c_m^2} C_m = 0, \quad (7.4a)$$

$$\frac{d}{dz} C_m(0) = \frac{d}{dz} C_m(-H) = 0 \quad (7.4b)$$

The eigenfunctions C_m are complete and orthogonal in the sense that

$$\int_{-H}^0 C_m C_n dz = \frac{c_m^2}{g} \delta_{mn}. \quad (7.5)$$

where $\delta_{mn} = 0$ unless $m = n$, in which case it equals one. The normalization is somewhat by convention and we include a factor of g for convenience to make the functions themselves nondimensional. There are an infinite number of eigenvalues, $c_m, c_0, c_1, c_2 \dots$, normally arranged in descending order, and for each there is a corresponding eigenfunction C_m . The pressure and horizontal velocity components are then expressed as

$$[u, v, \phi] = \sum_0^{\infty} [u_m(x, y, t), v_m(x, y, t), \phi_m(x, y, t)] C_m(z). \quad (7.6)$$

An practical advantage of this procedure is that the z -derivatives in the equations of motion are replaced by multiplications, and in particular (7.3) becomes

$$\frac{\partial \phi_m}{\partial t} + c_m^2 \nabla \cdot \mathbf{u}_m = 0. \quad (7.7a)$$

If we define $\eta^* = \phi/g$ then (7.7) becomes

$$\frac{\partial \eta_m^*}{\partial t} + H_m^* \nabla \cdot \mathbf{u}_m = 0. \quad (7.7b)$$

where $H_m^* = c_m^2/g$ is the *equivalent depth* associated with the eigenmode.

Equations (7.7) are evidently of the same form as the familiar linear mass continuity equation in the shallow water equations, which may be written

$$\frac{\partial \eta}{\partial t} + H \nabla \cdot \mathbf{u} = 0 \quad \text{or} \quad \frac{\partial \hat{\eta}}{\partial t} + c^2 \nabla \cdot \mathbf{u} = 0 \quad (7.8)$$

where $c = \sqrt{gH}$ and $\hat{\eta} = g\eta$.

The horizontal momentum equations are simply,

$$\frac{\partial u_m}{\partial t} - f v_m = -\frac{\partial \phi_m}{\partial x}, \quad \frac{\partial v_m}{\partial t} + f u_m = -\frac{\partial \phi_m}{\partial y}. \quad (7.9a,b)$$

Equations (7.8) and (7.9) are a closed set. If there is a forcing in the momentum equation then the transformed forcing appears on the right-hand sides of (7.9). If there is a source in the buoyancy equation then a corresponding term appears on the right-hand side of (7.8), analogous to a mass source term in the shallow water equations (see problem 7.??).

Eigenfunctions for vertical velocity

The vertical velocity and the buoyancy do not satisfy the same boundary conditions and so should not be expanded in the same way. Rather, we use eigenfunctions that satisfy the following relations.

$$\frac{1}{N^2} \frac{d^2 S_m}{dz^2} + \frac{1}{c_m^2} S_m = 0, \quad (7.10a)$$

$$S_m(0) = S_m(-H) = 0, \quad (7.10b)$$

where $S_m = 0$ if $N = 0$, and with the orthonormalization

$$\int_{-H}^0 N^2 S_m S_n dz = g \delta_{mn}. \quad (7.11)$$

These functions are related to C_m by

$$C_m = \frac{c_m^2}{g} \frac{dS_m}{dz} \quad N^2 S_m = -g \frac{dC_m}{dz}, \quad (7.12)$$

and $S_m = 0$ if $N = 0$.

Given the above, the vertical velocity may be evaluated from the mass continuity equation, $\partial w / \partial z = -\nabla \cdot \mathbf{u}$, which becomes

$$w_m \frac{dS_m}{dz} = -C_m \nabla \cdot \mathbf{u}_m \quad \Rightarrow \quad w_m = -\frac{c_m^2}{g} \nabla \cdot \mathbf{u}_m. \quad (7.13)$$

Approximations and interpretation

The values of c_m can be computed, in general, by solving the eigenvalue problem for the given stratification. In general this is a somewhat complex procedure that must be carried out numerically, but some approximate values can often be computed, especially if the stratification has certain simple forms.

First suppose N is constant. The normal modes are sines and cosines, and for $m = 0, 1, 2, \dots$ we have

$$C_m = A_m \cos \frac{m\pi z}{H}, \quad S_m = B_m \sin \frac{m\pi z}{H}, \quad c_m = \frac{NH}{m\pi}, \quad (7.14)$$

where, for $m > 0$, $A = c_m / \sqrt{gH/2}$ and $B = \sqrt{2g/HN^2}$. As an aside we note that the equivalent depth is then given by

$$H_e = \frac{N^2 h^2}{m^2 \pi^2 g} \sim \frac{g' H}{gm^2 \pi^2}, \quad (7.15)$$

where we define $g' \equiv H \partial b / \partial z = -(gH/\rho_0) \partial \rho / \partial z$.

The mode with $m = 0$ is a special one, and is called the *barotropic mode*. With N a constant we find

$$C_0 = A_0/2, \quad A_0 = \text{constant}, \quad c_0^2 = gH \quad (7.16)$$

and, perhaps most importantly, $c_0^2 = gH$. For an ocean of depth 5 km this gives $c_0 \approx 20 \text{ m s}^{-1}$, which is much higher than the velocity of fluid parcels or of the higher baroclinic modes. For $N = 10^{-2} \text{ s}^{-1}$ and $H = 1 \text{ km}$ (the scale of depth of the thermocline) we find $c_1 \approx 3 \text{ m s}^{-1}$, which is in agreement with numerical calculations that use a more realistic profile of stratification. (It is because the stratification is in reality concentrated in the upper ocean that such a value of H leads to a realistic answer.)

If the stratification varies sufficiently slowly, JWKB methods may be used to approximately evaluate the eigenvalues and eigenfunctions.² One may show (see the appendix) that

$$c_m \approx \frac{1}{m\pi} \int_{-H}^0 N dz \quad (7.17)$$

and

$$S_m \approx S_m^0 \sin \left(\frac{1}{c_m} \int_H^0 N(z) dz \right), \quad C_m \approx \left(\frac{c_m N S_m^0}{g} \right) \cos \left(\frac{1}{c_m} \int_{-H}^0 N(z) dz \right) \quad (7.18a,b)$$

If N is constant these reduce to the results obtained above; for N non-constant, the eigenfunctions are ‘stretched’ sines and cosines.

For much of the subsequent development in this chapter we will use the reduced-gravity shallow water form of the equations, rather than the normal mode form, because the notation is more familiar and the physical interpretation is a little simpler. However, this is a somewhat arbitrary choice, and it is always useful to remember that the equations have normal mode analogs that are valid for a continuously stratified ocean.

7.2 WAVES ON THE EQUATORIAL BETA PLANE

For small variations in latitude we use, as in section 3.2 of *AOFD*, the β -plane approximation: we Taylor-expand the Coriolis parameter around a latitude ϑ_0 and obtain

$$f = 2\Omega \sin \vartheta \approx 2\Omega \sin \vartheta_0 + 2\Omega (\vartheta - \vartheta_0) \cos \vartheta_0 = f_0 + \beta y \quad (7.19)$$

where $f_0 = 2\Omega \sin \vartheta_0$, $\beta = 2\Omega \cos \vartheta_0/a$ and $y = a(\vartheta - \vartheta_0)$. For motions at low latitudes we take $\vartheta_0 = 0$, giving the *equatorial beta-plane approximation* in which $\sin \vartheta \approx \vartheta$, $\cos \vartheta \approx 1$ and $f = 2\Omega \vartheta = \beta y$. The forced linearized momentum and mass conservation equations are then:

$$\frac{\partial u}{\partial t} - f v = -g' \frac{\partial \eta}{\partial x} + \frac{\tau^x}{H} \quad (7.20a)$$

$$\frac{\partial v}{\partial t} + f u = -g' \frac{\partial \eta}{\partial y} + \frac{\tau^y}{H} \quad (7.20b)$$

$$\frac{\partial \eta}{\partial t} + H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = M. \quad (7.20c)$$

where (τ^x, τ^y) is the kinematic wind stress (stress divided by seawater density) and M represents mass forcing, for example diffusion from the abyss. The stress term is often large, but the mass forcing is normally small. From (7.20) we may first derive the linearized potential vorticity equation, namely

$$\frac{\partial}{\partial t} \left(\zeta - \frac{f \eta}{H} \right) + \beta v = \frac{1}{H} \left(\frac{\partial \tau^y}{\partial x} - \frac{\partial \tau^x}{\partial y} - f M \right). \quad (7.21)$$

This is the same as the familiar linearized potential vorticity equation on the f -plane, with the addition of the term $Df/Dt = \beta v$. In all of the above equations, $f = \beta y$ and β is a constant.

To obtain a single equation for a single unknown, operate on (7.20a) with $(f/c^2)\partial_t$, on (7.20b) with $(1/c^2)\partial_{tt}$, on (7.20c) with $(g'/c^2)\partial_{ty}$ and on (7.21) with ∂_x , where $c^2 = g'H$. We suppose first that the wind- and mass-forcing are zero, and using subscripts to denote derivatives the resulting equations are,

$$\frac{f}{c^2} u_{tt} - \frac{f^2}{c^2} v_t = -\frac{f g'}{c^2} \eta_{xt}, \quad (7.22a)$$

$$\frac{1}{c^2} v_{ttt} + \frac{f}{c^2} u_{tt} = -\frac{g'}{c^2} \eta_{ytt}, \quad (7.22b)$$

$$\frac{g'}{c^2} \eta_{tty} + (u_{xyt} + v_{yyt}) = 0, \quad (7.22c)$$

$$v_{xxt} - u_{xyt} - \frac{g'f}{c^2} \eta_{xt} + \beta v_x = 0. \quad (7.22d)$$

These equations linearly combine to give a single equation for v , namely

$$\boxed{\frac{1}{c^2} \frac{\partial^3 v}{\partial t^3} + \frac{f^2}{c^2} \frac{\partial v}{\partial t} - \frac{\partial}{\partial t} \left(\frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial x^2} \right) - \beta \frac{\partial v}{\partial x} = 0.} \quad (7.23)$$

If f were constant and so $\beta = 0$ we could then straightforwardly obtain the dispersion relations

$$\omega = 0, \quad \omega^2 = f_0^2 + (k^2 + l^2)c^2. \quad (7.24a,b)$$

The first is the dispersion relation for geostrophic motion (which is stationary in the absence of a beta effect) and the second is the dispersion relation for Poincaré waves, previously obtained in section 3.7 of *AOFD*. When beta is non-zero the situation is considerably complicated, and we address that below. But before that we note that if τ and M are non-zero, potentially varying in both space and time, (7.23) becomes

$$\frac{1}{c^2} \frac{\partial^3 v}{\partial t^3} + \frac{f^2}{c^2} \frac{\partial v}{\partial t} - \frac{\partial}{\partial t} \left(\frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial x^2} \right) - \beta \frac{\partial v}{\partial x} = G. \quad (7.25a)$$

where

$$G = \frac{1}{H} \left[\frac{\partial}{\partial t} \left(\frac{1}{c^2} \frac{\partial \tau^y}{\partial t} - \frac{f}{c^2} \tau^x + \frac{\partial M}{\partial y} \right) - \frac{\partial}{\partial x} \left(\frac{\partial \tau^y}{\partial x} - \frac{\partial \tau^x}{\partial y} - fM \right) \right] \quad (7.25b)$$

Informal interpretation and mid-latitude analog

Although (7.25a) looks complicated, it is closely related to the familiar and simpler mid-latitude vorticity equation. Let us take f to be a constant, f_0 , except where differentiated, τ to be independent of time and $M = 0$. Cross differentiating (7.20a) and (7.20b) and using (7.20c) gives

$$\frac{\partial}{\partial t} \left(\zeta - \frac{f_0}{H} \eta \right) + \beta v = \text{curl}_z \tau \quad (7.26)$$

If the dominant balance in the momentum equations is geostrophic then the velocities are divergence-free and may be approximated by a streamfunction, $(u, v) \approx (-\psi_y, \psi_x)$, and $\eta \approx f_0 \psi / g'$. Equation (7.26) becomes

$$\frac{\partial}{\partial t} \left(\frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial x^2} - \frac{f^2}{c^2} \psi \right) + \beta \frac{\partial \psi}{\partial x} = \text{curl}_z \tau. \quad (7.27)$$

(A more systematic derivation is given in section 5.3.1 of *AOFD*.) We can see that (7.25a) is, aside from the third derivative in time, the x-derivative of (7.26).

7.3 WAVES

In this section we explore the properties of (7.23), in particular obtaining a dispersion relation. Our treatment is initially rather mathematical and formal, but we will follow this by a more physical discussion.

The coefficients of (7.23) vary in the meridional direction but are constant in the zonal direction. We thus search for solutions in the form of a plane wave in the zonal direction only and we let

$$v = \tilde{v}(y) e^{i(kx - \omega t)}, \quad (7.28)$$

and assume boundary conditions of $\tilde{v}(y) \rightarrow 0$ as $y \rightarrow \infty$. Substituting into (7.23) gives

$$\frac{d^2 \tilde{v}}{dy^2} + \left(\frac{\omega^2}{g'H} - k^2 - \frac{\beta k}{\omega} - \frac{\beta^2 y^2}{g'H} \right) \tilde{v} = 0. \quad (7.29)$$

Let us scale time and distance with the quantities

$$T_{eq} = (c\beta)^{-1/2}, \quad L_{eq} = (c/\beta)^{1/2} \quad (7.30a,b)$$

where $c \equiv \sqrt{g'H}$. The timescale T_{eq} is related to the lengthscale L_{eq} by $T_{eq} = (L_{eq}\beta)^{-1}$, and the non-dimensional frequency, lengthscale and wavenumber are given by

$$\hat{\omega} = \frac{\omega}{(\beta c)^{1/2}}, \quad \hat{y} = y \left(\frac{\beta}{c} \right)^{1/2}, \quad \hat{k} = k \left(\frac{c}{\beta} \right)^{1/2}. \quad (7.31)$$

The length scale L_{eq} is known as the *equatorial radius of deformation*, and it is a natural scale over which equatorial disturbances decay, as will become apparent very soon. If we take $\delta\rho/\rho_0 = 0.002$, $H = 100$ m and $\beta = 2\Omega/a = 2.3 \times 10^{-11}$ then we obtain

$$g' \approx 0.02 \text{ m s}^{-2}, \quad c \approx 1.4 \text{ m s}^{-1}, \quad L_{eq} \approx 250 \text{ km}, \quad T_{eq} = 1.7 \times 10^5 \text{ s} \approx 2 \text{ days}. \quad (7.32)$$

The shallow-water mid-latitude deformation radius, L_d is usually defined as $L_d = c/f$, which differs from (7.30b) most notably in the power of f . However, if in mid-latitude expression we take $f = \beta y$, as if near the equator, and $y = L_d$, then $L_d = c/(\beta L_d)$, which is the same as (7.30b).

Substituting (7.31) into (7.29) gives the simpler-looking equation

$$\frac{d^2 v}{d\hat{y}^2} + \left(\hat{\omega}^2 - \hat{k}^2 - \frac{\hat{k}}{\hat{\omega}} - \hat{y}^2 \right) v = 0. \quad (7.33)$$

This equation may be put into a standard form³ by writing

$$v(\hat{y}) = \Psi(\hat{y}) e^{-\hat{y}^2/2}, \quad (7.34)$$

whence (7.33) becomes

$$\frac{d^2 \Psi}{d\hat{y}^2} - 2\hat{y} \frac{d\Psi}{d\hat{y}} + \lambda \Psi = 0 \quad (7.35)$$

where $\lambda = \hat{\omega}^2 - \hat{k}^2 - \hat{k}/\hat{\omega} - 1$. Equation (7.35) is known as *Hermite's equation*, and it is an eigenvalue equation, with solutions if and only if $\lambda = 2m$, for $m = 0, 1, 2, \dots$. The solutions are Hermite polynomials, $\Psi(\hat{y}) = H_m(\hat{y})$, where the first few polynomials are given by

$$H_0 = 1, \quad H_1 = 2\hat{y}, \quad H_2 = 4\hat{y}^2 - 2, \quad (7.36)$$

$$H_3 = 8\hat{y}^3 - 12\hat{y}, \quad H_4 = 16\hat{y}^4 - 48\hat{y}^2 + 12. \quad (7.37)$$

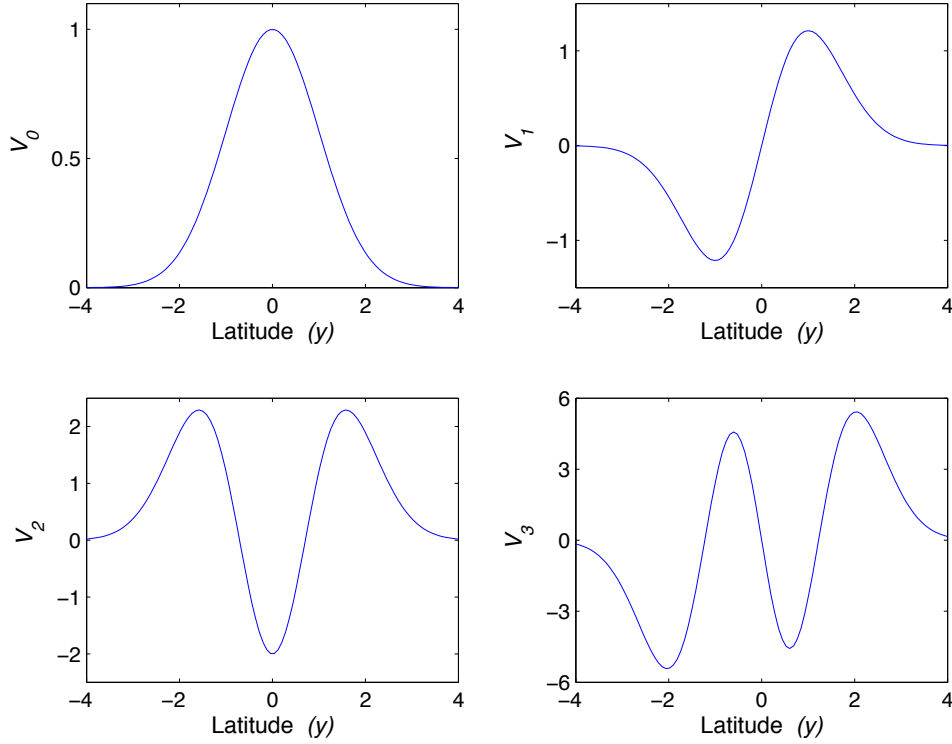


Fig. 7.1 Latitudinal variation of the wave amplitudes, $V_m(y)$, given by (7.39) as a function of non-dimensional latitude, y for $m = 0, 1, 2, 3$. The parameter m is like a meridional wavenumber.

A Hermite polynomial is even or odd when m is even or odd, respectively; that is $H_m(-\hat{y}) = (-1)^m H_m(\hat{y})$. The polynomials are also orthogonal in the interval $[-\infty, +\infty]$; that is

$$\int_{-\infty}^{\infty} H_n H_m = C_n \delta_{mn} \quad (7.38)$$

where $C_n = 2^n n! \pi^{1/2}$ is a normalization constant. The solutions for v are the Hermite functions, multiplied by the Gaussian centered at the equator, so that

$$v(y) = V_m(y) = H_m e^{-\hat{y}^2/2}, \quad m = 0, 1, 2, \dots \quad (7.39)$$

and so decay exponentially as $y \rightarrow \pm\infty$ (as we require) with a decay scale of the equatorial deformation radius $\sqrt{c/\beta}$. The functions V_m are plotted in Fig. 7.1 for $m = 0$ to 3.

The dispersion relation follows from the quantization condition $\lambda = 2m$, which implies

$$\hat{\omega}^2 - \hat{k}^2 - \frac{\hat{k}}{\hat{\omega}} = 2m + 1 \quad \text{or} \quad \omega^2 - c^2 k^2 - \beta \frac{kc^2}{\omega} = (2m + 1)\beta c, \quad (7.40a,b)$$

using (7.31) to obtain the dimensional expression on the right. This is a cubic equation in ω , and although a general solution for $\hat{\omega}$ is possible, it is easier to solve the quadratic

equation for $\hat{k}$ in terms of $\hat{\omega}$ giving, after a little algebra,

$$\hat{k} = -\frac{1}{2\hat{\omega}} \pm \frac{1}{2} \left[\left(\frac{1}{\hat{\omega}} - 2\hat{\omega} \right)^2 - 8m \right]^{1/2}, \quad (7.41a)$$

or, in dimensional form,

$$k = -\frac{\beta}{2\omega} \pm \frac{1}{2} \left[\left(\frac{\beta}{\omega} - \frac{2\omega}{c} \right)^2 - \frac{8m\beta}{c} \right]^{1/2}. \quad (7.41b)$$

7.3.1 Limiting and special cases

To understand more physically what is going on we may use (7.40) and (7.41) to consider various special cases. It is convenient to first partition the waves by frequency, and consider separately high frequency gravity waves and low frequency planetary waves. We need do this only for the case $m \geq 1$ because the $m = 0$ case (mixed Rossby-gravity waves) may be treated exactly. Then finally we look at the so-called $m = -1$ case, namely Kelvin waves.

- (i) *High frequency waves.* The term $\beta kc^2/\omega$ in (7.40) is small and may be neglected. The dispersion relation becomes

$$\hat{\omega}^2 = \hat{k}^2 + 2m + 1 \quad \text{or} \quad \omega^2 = c^2 k^2 + \beta c(2m + 1). \quad (7.42a,b)$$

This dispersion relation is similar to that of mid-latitude Poincaré waves, with βc replacing f_0^2 : recall the form of (3.105) of AOFD, namely $\omega^2 = c^2(k^2 + l^2) + f_0^2$. Waves satisfying (7.42) are thus sometimes called equatorially trapped Poincaré waves or equatorially trapped gravity waves.

The approximation requires that $\omega \gg \beta/|k|$, and is somewhat inaccurate for small k : note that (7.42) is symmetric around $k = 0$, whereas the full dispersion relation, plotted in Fig. ??, is offset. (Formally, the limit is valid for $\hat{k} \rightarrow \infty$, $\hat{\omega} \rightarrow \infty$ and $k/\hat{\omega} = \text{constant}$.)

For finite m the limiting case at high wavenumber just $\omega = \pm \hat{k}$, or, in dimensional form, $\omega = \pm ck$. This is just the dispersion relation for familiar conventional shallow water gravity waves, unaffected by rotation and the β -effect. However, in the rotating case the waves are trapped at the equator and propagate only in the zonal direction, albeit both eastward and westward.

- (ii) *Low frequency waves.* For low frequency waves we neglect the term involving ω^2 in (7.40) and the dispersion relation becomes

$$\hat{\omega} = \frac{-\hat{k}}{2m + 1 + \hat{k}^2} \approx -\frac{1}{\hat{k}}, \quad \omega = \frac{-\beta k}{(2m + 1)\beta/c + k^2}, \quad (7.43)$$

non-dimensionally and dimensionally, respectively. This is recognizable as the dispersion relation for a zonally propagating Rossby wave with large x-wavenumber, and these waves are called equatorially trapped Rossby waves, or equatorially trapped planetary waves. We may further consider two limits of these waves, as follows.

(a) *Short, low frequency waves, with $\hat{k} \rightarrow \infty$, $\hat{\omega} \rightarrow 0$.* The dispersion relation becomes

$$\omega = \frac{1}{\hat{k}}, \quad \omega = -\frac{\beta}{k}. \quad (7.44)$$

The phase speed and group velocity in this limit are given by, dimensionally,

$$c_p = -\frac{\beta}{k^2}, \quad c_g = \frac{\beta}{k^2}, \quad (7.45)$$

Thus, the phase speed is westward but the group velocity, and so the direction of energy propagation, is eastward.

(b) *Long low frequency waves, with $\hat{k} \rightarrow 0$, $\hat{\omega} \rightarrow 0$.* The dispersion relation (7.40) becomes, in nondimensional and dimensional form,

$$\hat{\omega} = \frac{-\hat{k}}{2m+1}, \quad \omega = \frac{-ck}{2m+1}, \quad (7.46)$$

These represent westward propagating waves whose speed is given by $c/(2m+1)$, similar to that of a gravity wave. However, like planetary waves they propagate only westward, and they match with the westward propagating planetary waves derived above as wavenumber increases.

There is a distinct gap in frequencies between the minimum frequency of the gravity waves, given by (7.42), and the maximum frequency of the planetary waves, given by (7.45) also with m small. The minimum gravity wave frequency occurs when $k = 0$ and is $\omega_{gmin}^2 = \beta c(2m+1)$. From (7.44) the maximum planetary wave frequency occurs when $k^2 = (2m+1)\beta/c$ and gives $\omega_{pmax}^2 = \beta c/[4(2m+1)]$. The ratio of these two frequencies is

$$\frac{\omega_{gmin}}{\omega_{pmax}} = 2(2m+1), \quad (7.47)$$

giving a value of six for $m = 1$ and two for $m = 0$ (a case we consider more below). Note that this ratio is *independent* of the values of the physical parameters β and c . Although the gap is distinct, it is not as large as the corresponding gap at midlatitudes, which may be an order of magnitude or more.

Special values of m

In addition to consider limiting cases at low and high frequency, there are two other cases in which we can solve the dispersion relation, namely the case with $m = 0$ and the Kelvin wave case, as follows.

(i) *The case with $m = 0$.* The resulting waves are known as *Yanai waves*,⁴ From (7.41a) dispersion relation simplifies to the two cases

$$\hat{k} = -\hat{\omega}, \quad \hat{k} = -\frac{1}{\hat{\omega}} + \hat{\omega}. \quad (7.48a,b)$$

or dimensionally

$$k = -\frac{\omega}{c}, \quad k = -\frac{\beta}{\omega} + \frac{\omega}{c}. \quad (7.49a,b)$$

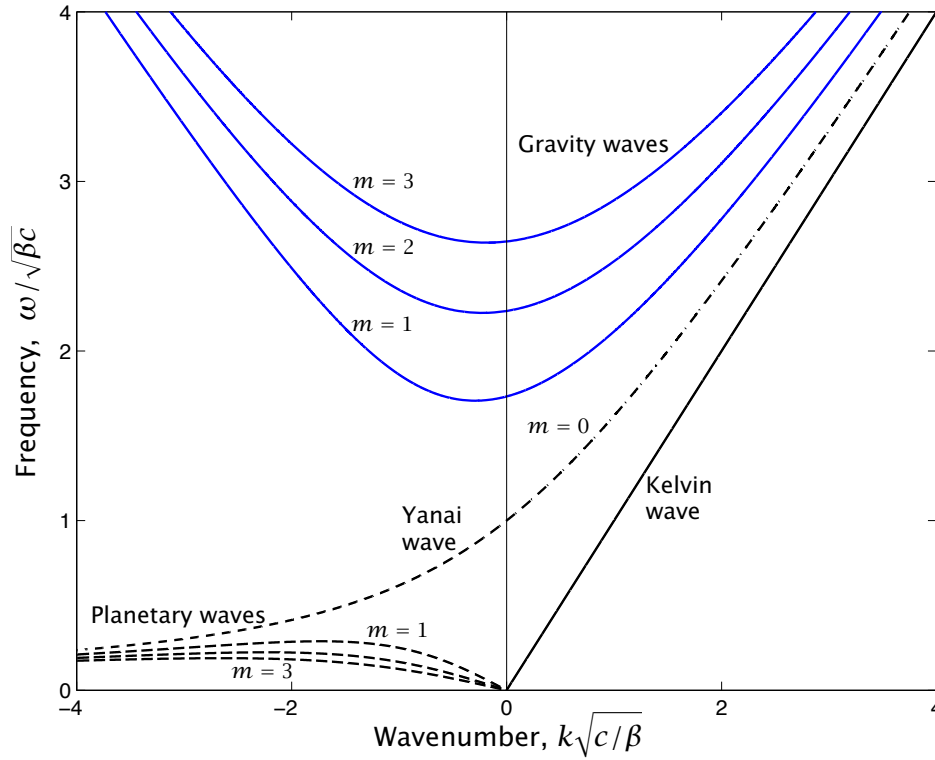


Fig. 7.2 Dispersion relation for equatorial waves, as given by (7.40), for $m = 0, 1, 2, 3$. The upper group of curves are gravity waves, given approximately by (7.42). The lower group with $k < 0$ are westward propagating planetary waves, given approximately by (7.43). Also shown are the Yanai wave with $m = 0$, satisfying (7.50), and the eastward propagating Kelvin wave (the ' $m = -1$ ' wave) satisfying $\omega = ck$ for $k \geq 0$.

The case $k = -\omega/c$ is actually spurious, although this is not obvious. [Need explanation! xx]. The physically realizable case, (7.49b) has the explicit dispersion relation

$$\omega = \frac{kc}{2} \pm \frac{1}{2} \sqrt{k^2 c^2 + 4\beta c}. \quad (7.50)$$

Again it is useful to consider limiting cases, as follows.

- $k = 0$. In this case (7.50) gives $\omega = \sqrt{\beta c}$ and there is a balance between two terms on the right-hand side of (7.50).
- $k \rightarrow +\infty$. In this case $\omega = ck$, with a balance between the left-hand side and the second term on the right-side of (7.49b). These are eastward propagating gravity waves.
- $k \rightarrow -\infty$. In this case, because ω must be positive, we have $\omega = -\beta/k$, and a balance between the left-hand side and the first term on the right-side of (7.49b).

Yanai waves, therefore, are mixed Rossby-gravity waves: the phase of the Rossby wave propagates westward (like all Rossby waves) and has a low frequency, and the gravity wave propagates eastward (and only eastward, unlike conventional gravity waves). The group velocity of Yanai waves is positive in all cases, being given by, from (7.49b),

$$c_g \equiv \frac{\partial \omega}{\partial k} = \frac{\omega^2 c}{\beta c + \omega^2}. \quad (7.51)$$

The group velocity of the full problem is, from (7.40),

$$c_g = \frac{c^2 \omega (\beta + 2\omega k)}{2\omega^3 + \beta k c^2}. \quad (7.52)$$

This may be positive or negative, and vanishes when $\omega = -\beta/(2k)$.

- (ii) *Kelvin waves, or the 'm = -1' case.* Hermite's equation, (7.35) has solutions when m is a positive integer or zero. However, there is a class of waves that satisfies the dispersion relation (7.40) with $m = -1$, namely equatorial Kelvin waves, as we shall now discover. (This section follows on from 3.7.3 of AOFD, and the reader may wish to peruse that section before continuing.)

Kelvin waves have identically zero meridional velocity and so their equations of motion are

$$\frac{\partial u}{\partial t} = -g' \frac{\partial \eta}{\partial x}, \quad f u = -g' \frac{\partial \eta}{\partial y}, \quad \frac{\partial \eta}{\partial t} + H \frac{\partial u}{\partial x} = 0. \quad (7.53a,b,c)$$

where $f = \beta y$. The zonal velocity is in geostrophic balance with the meridional pressure gradient, and (7.53a) and (7.53c) give the classic wave equation,

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0. \quad (7.54)$$

where $c = \sqrt{g'H}$ as before, and so the dispersion relation $\omega = ck$. This is, in fact, a solution of (7.40) with $m = -1$, as may easily be checked.

The solution to (7.54), and the corresponding solution for η , is

$$u = F_1(x + ct, y) + F_2(x - ct, y), \quad \eta = \sqrt{\frac{H}{g'}} [-F_1(x + ct, y) + F_2(x - ct, y)] \quad (7.55)$$

where F_1 and F_2 are arbitrary functions, representing waves travelling westwards and eastwards, respectively. We obtain the y -dependence of these functions by using (7.53b) giving

$$\beta y F_1 = c \frac{\partial F_1}{\partial y}, \quad \beta y F_2 = -c \frac{\partial F_2}{\partial y}. \quad (7.56)$$

The solutions of these equations are

$$F_1 = F(x + ct) \exp[y^2/(2L_{eq}^2)], \quad F_2 = G(x - ct) \exp[-y^2/(2L_{eq}^2)] \quad (7.57a,b)$$

where F and G are the amplitudes at $y = 0$. Evidently, F_1 increases without bound away from the equator, and so this solution must be eliminated. The complete solution is thus:

$$u = G(x - ct) \exp[-y^2/(2L_{eq}^2)], \quad \eta = \frac{H}{c} u, \quad v = 0. \quad (7.58)$$

These waves are equatorially trapped Kelvin waves. They propagate eastward only, without dispersion, and their amplitude decays away from the equator in precisely the same way as the other equatorial waves considered above.

7.4 RAY TRACING AND EQUATORIAL TRAPPING

We have seen that equatorial waves are trapped near the equator. What then happens to a wave that initially propagates in a direction away from the equator? The waves must either change their character completely, or be refracted back toward the equator. The former can only happen if there exists a class of midlatitude waves with similar frequency and wavenumber; otherwise no such waves can be excited and the waves must, if they are not absorbed, bend back if energy is to be conserved. Let us explore this using some ideas from ray theory (see the appendix to chapter 5 and section 13.2 of *AOFD*).

7.4.1 Dispersion relation and ray equations

Consider again the wave equation of motion for the meridional velocity, namely (7.29) or

$$\frac{d^2 \tilde{v}}{dy^2} + \left(\frac{\omega^2}{c^2} - k^2 - \frac{\beta k}{\omega} - \frac{\beta y^2}{c^2} \right) \tilde{v} = 0. \quad (7.59)$$

If the term in brackets is positive then sinusoidal solutions in y are possible, but if the term is negative, which will occur for y larger than some critical value y_c , then the physically realizable solutions decay exponentially with y . The wavelike solutions are therefore trapped between two critical latitudes given by

$$y_c = \pm \left(\frac{\omega^2}{\beta^2} - \frac{c^2 k^2}{\beta^2} - \frac{c^2 k}{\beta \omega} \right)^{1/2} = \left((2m+1) \frac{c}{\beta} \right)^{1/2}, \quad (7.60)$$

using the dispersion relation to obtain the rightmost expression. For $k = 0$, and so for meridionally propagating waves, the critical latitudes are given by $y_c = \omega/\beta$, and so at the critical latitude $\omega = f$. The waves are therefore trapped within their *inertial latitudes*, the latitudes at which their frequency is f . For larger k the critical latitudes are correspondingly smaller.

To explore this phenomenon using ray theory we assume that the medium is varying sufficiently slowly that it is possible to find wavelike solutions with spatially varying wavenumbers. Here the medium varies only in the y -direction (because of the β effect) and so we seek solutions of the form

$$v = V(y) e^{i(kx + l(y)y - \omega t)}, \quad (7.61)$$

where we assume that $V(y)$ varies slowly enough with y so that its derivatives are small. Equation (7.61) is the same as letting $\tilde{v}(y) = V(y) e^{il(y)y}$ in (7.59), which then gives an equation for l ,

$$l^2 = \frac{\omega^2}{c^2} - k^2 - \frac{\beta k}{\omega} - \frac{\beta y^2}{c^2} = \frac{\beta^2}{c^2} (y_c^2 - y^2). \quad (7.62)$$

Writing (7.59) as

$$\frac{d^2 V}{dy^2} + l^2(y) V = 0, \quad (7.63)$$

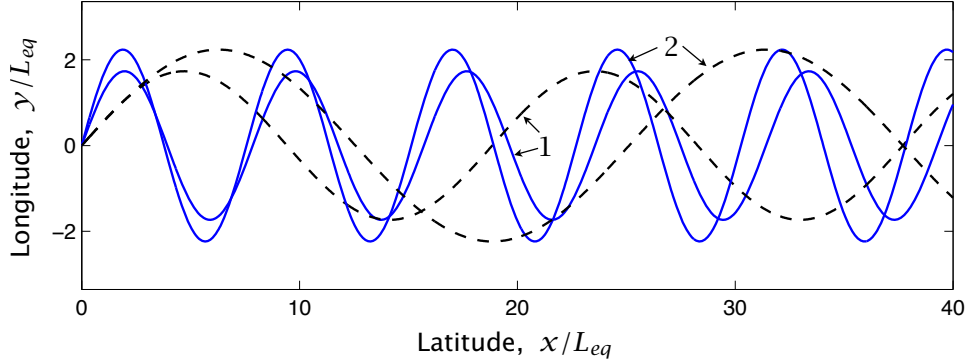


Fig. 7.3 Rays in the equatorial waveguide calculated using (7.70). The dashed lines show planetary wave trajectories and the solid lines are gravity wave trajectories, with $m = 1, 2$ (numbers marked on the graph) and $\hat{k} = 1$. The turning latitude for each wave is $(2m + 1)^{1/2}L_{eq}$, where $L_{eq} = \sqrt{c/\beta}$.

the approximate solution, as given by WKBJ methods (see Appendix), is

$$V = l^{-1/2} \exp\left(\pm i \int l \, dy\right). \quad (7.64)$$

The formal condition for the validity of this solution is

$$\frac{d^2 l^{-1/2}}{dy^2} \ll l^{3/2}, \quad (7.65)$$

which roughly speaking is that the meridional scale of the wave, l^{-1} should be small compared to the scale over which l varies. [Expand this in brief appendix.]

Wave packets travel along rays — paths that are parallel to the direction of the group velocity. That is, their trajectory, $x(t), y(t)$ is defined by

$$\frac{dx}{dt} = c_g^x, \quad \frac{dy}{dt} = c_g^y, \quad \text{so that} \quad \frac{dy}{dx} = \frac{c_g^y}{c_g^x}. \quad (7.66)$$

Using the dispersion relation (7.62) gives

$$\frac{\partial \omega}{\partial l} = \frac{2\omega^2 l c^2}{2\omega^3 + \beta k c^2}, \quad (7.67)$$

and using this and (7.52) gives the slope of the ray in the x - y plane,

$$\frac{dy}{dx} = \frac{c_g^y}{c_g^x} = \frac{l}{k + \beta/(2\omega)}. \quad (7.68)$$

Using (7.62) we can write this in terms of y instead of l , so that

$$\frac{dy}{dx} = \frac{\beta(y_c^2 - y^2)^{1/2}}{kc + \beta c/(2\omega)}. \quad (7.69)$$

Using the standard result that

$$\int \frac{dy}{(y_c^2 - y^2)^{1/2}} = \sin^{-1} y/y_c \quad (7.70)$$

we finally obtain

$$y = y_c \sin \left[\frac{\beta x}{ck + \beta c/(2\omega)} \right], \quad \hat{y} = (2m + 1) \sin \left[\frac{\hat{x}}{\hat{k} + 1/(2\hat{\omega})} \right]. \quad (7.71)$$

where the second expression is the nondimensional form. The ray path is therefore sinusoid moving along the equator; the waves are confined to a *waveguide* centered at the equator and with a polewards extent of $y = \pm y_c$. Equation (7.71) holds for both planetary and gravity waves, and for the latter the term $\beta c/(2\omega)$ may be neglected.

7.4.2 Discussion

Some observations and implications.

7.5 FORCED MOTION NEAR THE EQUATOR

Notes

- 1 One could write a whole book on this, and some people have.
- 2 See Chelton *et al.* (1998) for a description of the method and maps of the first deformation radius and related quantities for the world's oceans.
- 3 Standard forms are in the eye of the beholder.
- 4 After Yanai. Need some background here. xxx